

Polynomials (Part I): A Review of Polynomials Homework

Determine if the expressions below are polynomials. If not, circle the term(s) that disqualify each expression.

- ① $3x+4$
- ② $8x^5 - \pi x + 16$
- ③ $-10 + x^{-7} + 4x$
- ④ $x^5 + 3x^4 - 8x^3 + x^2 - 2$
- ⑤ $721.3x^4 - \frac{22}{3}x^8 + x$
- ⑥ $\sqrt[3]{x^5} - 8x^2 + 7x - 9$
- ⑦ $2x^2 + \frac{1}{x} - 4x - 13$
- ⑧ $43x^3 + 88x^2 + \frac{x}{6} + 22$
- ⑨ $8x^3 + 2x^5 - \frac{1}{8}x$
- ⑩ $37x^{19} - 5x^{-3} + \pi x^4$

Simplify the following polynomials, write in descending order of powers, and state the degree.

- ⑪ $5x^6 + 2x^2 - 3x^6 + 6x - 7x^2 + 2$
- ⑫ $-5x - 2x^3 + 10 + 3x^2 - 2x - 9 + 8x^3 - 2x + 5x + 10x^2$
- ⑬ $2 + x^2 - 8x^2 + 3x^2 + 6x + 4 - 8x + x^2 - 5x^5$

$$\textcircled{14} -x^3 + 2x - 4x + 8 - 3 + 2x - 1$$

Add or subtract the following polynomials.

- ⑮ $(2x^3 - 3x^2 + 7x + 5) + (5x^2 - 6x + 2)$
- ⑯ $(12x^3 - 6x^2 + 7x + 5) - (5x^4 - x^3 + 2x^2 - 2x - 1)$
- ⑰ $(x^3 - 3x^2 + 7x + 5) + (9x + 2)$
- ⑱ $(-x^3 + 3x) - (-8x^3 + 9x + 2)$
- ⑲ $(4x - 6) + (-3x^2 + 9x - 1)$
- ⑳ $(7x^3 + 4x^2 - 2x + 5) - (3x^2 - 4x + 8)$
- ㉑ $(-8x^3 + x^2) + (-4x^3 - 8x^2 - 7x + 13)$

Multiply.

- ㉒ $4x^3(2x^2 - 3x + 9)$
- ㉓ $y^3(2y^3 + 3y^2 - 4y + 6)$
- ㉔ $-z^8(2z^3 + 6z^2 - z + 6)$
- ㉕ $3x(4x^2 - 2x + 10)$
- ㉖ $-10x^3(7x^3 - 7x^2 + 2)$
- ㉗ $(x+9)(x+8)$
- ㉘ $(2x-3)(2x-5)$

$$(29) (x+5)(9x^2+x-3)$$

$$(30) (x^2+4)(-3x^2+4x+1)$$

Divide.

$$(31) \frac{20x^5 - 35x^4 + 10x^3 - 30x^2}{5x}$$

$$(32) \frac{64x^3 - 40x^2}{8x^2}$$

$$(33) \frac{100x^9 + 50x^5 - 80x^3 + 20x^2}{-10x}$$

$$(34) \frac{15x^2 + 3x + 9}{3}$$

Perform the following operations.

$$(35) 774 \div 6$$

$$(36) 3,696 \div 7$$

$$(37) 949 \div 4$$

$$(38) 2,897 \div 5$$

Answers

- ① $3x+4$ Yes
- ② $8x^5 - \pi^x + 16$ No
- ③ $-10 + x^{-7} + 4x$ No
- ④ $x^5 + 3x^4 - 8x^3 + x^2 - 2$ Yes
- ⑤ $721.3x^4 - \frac{23}{3}x^2 + x$ Yes
- ⑥ $\sqrt[3]{x^5} - 8x^2 + 7x - 9$ No
- ⑦ $2x^2 + \frac{1}{x} - 4x - 13$ No
- ⑧ $43x^3 + 28x^2 + \frac{x}{6} + 122$ Yes
- ⑨ $8x^0 + 2x^5 - \frac{1}{8}x$ Yes
- ⑩ $37x^{19} - 5^{x-3} + \pi x^4$ No
- ⑪ $2x^6 - 5x^2 + 6x + 2, D=6$
- ⑫ $6x^3 + 13x^2 - 4x + 1, D=3$
- ⑬ $-5x^5 - 3x^2 - 2x + 6, D=5$
- ⑭ $-x^3 + 4, D=3$
- ⑮ $2x^3 + 2x^2 + x + 7$
- ⑯ $-5x^4 + 13x^3 - 8x^2 + 9x + 6$
- ⑰ $x^3 - 3x^2 + 16x + 7$

- ⑱ $7x^3 - 6x - 2$
- ⑲ $-3x^2 + 13x - 7$
- ⑳ $7x^3 + x^2 + 2x - 3$
- ㉑ $-12x^3 - 7x^2 - 7x + 13$
- ㉒ $8x^5 - 12x^4 + 36x^3$
- ㉓ $2y^8 + 3y^7 - 4y^6 + 6y^5$
- ㉔ $-2z^{11} - 6z^{10} + z^9 - 6z^8$
- ㉕ $12x^3 - 6x^2 + 30x$
- ㉖ $-70x^6 + 70x^5 - 20x^3$
- ㉗ $x^2 + 17x + 72$
- ㉘ $4x^6 - 16x + 15$
- ㉙ $9x^3 + 46x^2 + 2x - 15$
- ㉚ $-3x^4 + 4x^3 - 11x^2 + 16x + 4$
- ㉛ $4x^4 - 7x^3 + 2x^2 - 6x$
- ㉜ $8x - 5$
- ㉝ $-10x^8 - 5x^4 + 8x^2 - 2x$
- ㉞ $5x^2 + x + 3$
- ㉟ 129
- ㊱ 528
- ㊲ $237 + \frac{1}{4}$
- ㊳ $579 + \frac{2}{5}$